

Rank-One Composition Invariance and Reflexive BAP Targets for Positively Homogeneous Free Banach Lattice Function Spaces

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Status

This is a strengthened partial result for Laustsen and Tradacete [1]. First, it shows that their printed Question 6.3 is not an independent obstruction: for each target space F , the full composition-invariance statement of Question 6.3 is equivalent to the equality in Question 4.2 for F . Moreover, failure of Question 4.2 is already detected by rank-one positively homogeneous maps. Second, it gives a broad positive target class for Question 4.2: every reflexive real Banach space with the bounded approximation property has, for every $1 \leq p < \infty$,

$$\overline{I_{w^*}[E]} = H_{w^*}^p[E].$$

Consequently Question 6.3 is affirmative for all such targets.

Source Questions

The source paper introduces, for a Banach space E and $1 \leq p < \infty$, the Banach lattice $H^p[E]$ of positively homogeneous functions on E^* , the weak-star continuous sublattice $H_{w^*}^p[E]$, the ideal $I_{w^*}[E]$, and its closure $\overline{I_{w^*}[E]}$ in the $\text{FBL}^p[E]$ -norm. It asks:

4. $H_{w^*}^p$ VS. FBL^p FOR INFINITE-DIMENSIONAL SPACES

Theorem 3.5 shows that when the Banach space E is infinite-dimensional, we should replace $I[E]$ and $I_{w^*}[E]$ with their closures $\overline{I[E]}$ and $\overline{I_{w^*}[E]}$ in the norm $\|\cdot\|_{\text{FBL}^p[E]}$, for $1 \leq p < \infty$, before we compare them with the other Banach lattices we study. In this section we shall focus on the inclusions

$$\text{FBL}^p[E] \subseteq \overline{I_{w^*}[E]} \subseteq H_{w^*}^p[E].$$

We note that all three sets are closed sublattices of $H^p[E]$, and Theorem 3.5 shows that $\text{FBL}^p[E] = H_{w^*}^p[E]$ when E is finite-dimensional. However, at this point we do not know whether one or both of these inclusions could be an equality for some infinite-dimensional Banach space E , raising the following two questions:

Question 4.1. For which Banach spaces E is it true that $\text{FBL}^p[E] = \overline{I_{w^*}[E]}$ for some/all $1 \leq p < \infty$?

Question 4.2. For which Banach spaces E is it true that $\overline{I_{w^*}[E]} = H_{w^*}^p[E]$ for some/all $1 \leq p < \infty$?

We begin by addressing the first of these questions. Our aim is to prove the following result.

Theorem 4.3. *Let E be a Banach space which admits an infinite-dimensional, separable quotient space, and let $1 \leq p < \infty$. Then $I_{w^*}[E] \not\subseteq \text{FBL}^p[E]$, and hence*

$$\text{FBL}^p[E] \subsetneq \overline{I_{w^*}[E]}.$$

We remark that no infinite-dimensional Banach space E which fails to admit an infinite-dimensional, separable quotient space is known. In view of this, we conjecture that the condition that $\text{FBL}^p[E] = \overline{I_{w^*}[E]}$ for some (or all) $1 < p < \infty$

Later, after proving that every lattice homomorphism between free p -convex Banach lattices is a composition operator and giving rank-one examples that do not preserve FBL^p , the paper asks:

Example 6.2. Identify ℓ_∞ with ℓ_1 , as usual. By [20] Remark 10.6], there is a positively homogeneous map $\Phi: \ell_\infty \rightarrow \ell_\infty$ satisfying the hypotheses of Lemma 6.1(ii) for $p = 1$, that is, $\|\Phi\|_1 < \infty$ and $\Phi|_{B_{\ell_\infty}}$ is weak*-to-weak* continuous, but the corresponding composition operator C_Φ does not map $\text{FBL}^1[\ell_1]$ into itself.

We can now provide a much more general version of this result. Take $1 \leq p < \infty$, and let E and F be Banach spaces, where E is non-zero and F admits an infinite-dimensional, separable quotient space. By Theorem 4.3 we can find a function $f \in H_{w^*}^p[F] \setminus \text{FBL}^p[F]$. (In fact, we can take $f \in I_{w^*}^p[F] \setminus \text{FBL}^p[F]$, but this will not help the following argument.) Choose $x_0^* \in E^* \setminus \{0\}$, and define a map $\Phi: F^* \rightarrow E^*$ by $\Phi(y^*) = f(y^*)x_0^*$ for $y^* \in F^*$. Since $f \in H_{w^*}^p[F]$, it is straightforward to check that Φ is positively homogeneous, $\Phi|_{B_{F^*}}$ is weak*-to-weak* continuous, and $\|\Phi\|_p = \|f\|_{\text{FBL}^p[F]} \|x_0^*\| < \infty$. However, if we pick $x_0 \in E$ such that $x_0^*(x_0) = 1$, then

$$C_\Phi(\delta_{x_0})(y^*) = (\delta_{x_0} \circ \Phi)(y^*) = \Phi(y^*)(x_0) = f(y^*) \quad (y^* \in F^*),$$

so $C_\Phi(\delta_{x_0}) = f \notin \text{FBL}^p[F]$. Hence C_Φ does not map $\text{FBL}^p[E]$ into $\text{FBL}^p[F]$.

The following question arises naturally from the inclusions (6.2)–(6.3) and Example 6.2. We have not been able to answer it, partly due to the lack of progress on Question 4.2

Question 6.3. Let E and F be infinite-dimensional Banach spaces, and suppose that $\Phi: F^* \rightarrow E^*$ is a positively homogeneous map which satisfies $\|\Phi\|_p < \infty$ for some $1 \leq p < \infty$ and $\Phi|_{B_{F^*}}$ is weak*-to-weak* continuous. Is

$$C_\Phi(\overline{I_{w^*}[E]}) \subseteq \overline{I_{w^*}[F]}?$$

We shall now present our main representation theorem for lattice homomorphisms between free Banach lattices

We write $C_\Phi f = f \circ \Phi$.

Proof Intuition

Question 6.3 asks whether every admissible composition operator preserves the closed ideal $\overline{I_{w^*}}$. If Question 4.2 is true for the target F , then this is immediate: the source paper already proves that such a C_Φ sends weak-star continuous H^p -functions to weak-star continuous H^p -functions. Therefore the image lands in $H_{w^*}^p[F] = \overline{I_{w^*}[F]}$.

Conversely, there is no need to search among complicated nonlinear maps. Given any $f \in H_{w^*}^p[F]$, choose a nonzero coordinate $x_0^* \in E^*$ and set $\Phi_f(y^*) = f(y^*)x_0^*$. Then Φ_f satisfies exactly the hypotheses of Question 6.3, and composition sends a single generator $\delta_{x_0} \in I_{w^*}[E]$ to f . Hence Question 6.3 for the fixed target F forces every $f \in H_{w^*}^p[F]$ to belong to $\overline{I_{w^*}[F]}$, which is precisely Question 4.2 for F .

For reflexive spaces with the bounded approximation property, the representation theorem used in the source paper can be combined with finite-rank approximation on the support of the representing measure. A positive measure μ on the weakly compact unit ball defines

$$f_\mu^p(x^*) = \left(\int |x^*(x)|^p d\mu(x) \right)^{1/p}.$$

The weakly compact ball is an Eberlein compact, so the Amir–Lindenstrauss separable-support theorem for Radon measures on Eberlein compacta lets us assume μ is carried by a separable

closed subspace $S \subseteq E$. The bounded approximation property gives a uniformly bounded finite-rank sequence T_n converging strongly to the identity on S . Replacing x by $T_n x$ gives a function depending only on a finite-dimensional subspace. By the finite-dimensional equality in the source paper, this compressed function belongs to $\text{FBL}^p[E]$, hence to $\overline{I_{w^*}[E]}$. The FBL^p -error is controlled by

$$\left(\int \| (I - T_n)x \|^p d\mu(x) \right)^{1/p},$$

which tends to zero by dominated convergence.

Main Result

Theorem 1 (Target-space equivalence). *Fix $1 \leq p < \infty$ and an infinite-dimensional Banach space F . The following are equivalent.*

1. $\overline{I_{w^*}[F]} = H_{w^*}^p[F]$.
2. For every infinite-dimensional Banach space E , and every positively homogeneous map $\Phi : F^* \rightarrow E^*$ such that $\|\Phi\|_p < \infty$ and $\Phi|_{B_{F^*}}$ is weak-star to weak-star continuous,

$$C_\Phi(\overline{I_{w^*}[E]}) \subseteq \overline{I_{w^*}[F]}.$$

3. For one infinite-dimensional Banach space E , one vector $x_0 \in E$, and one functional $x_0^* \in E^*$ with $x_0^*(x_0) = 1$, the following rank-one test holds: for every $f \in H_{w^*}^p[F]$, the map $\Phi_f : F^* \rightarrow E^*$, $\Phi_f(y^*) = f(y^*)x_0^*$, satisfies

$$C_{\Phi_f}(\delta_{x_0}) \in \overline{I_{w^*}[F]}.$$

Proof. (1) $\Rightarrow$ (2). Let $g \in \overline{I_{w^*}[E]}$. By Lemma 2.1(viii) of [1], $\overline{I_{w^*}[E]} \subseteq H_{w^*}^p[E]$. Lemma 6.1(ii) of [1] says that the hypotheses on Φ imply

$$C_\Phi(H_{w^*}^p[E]) \subseteq H_{w^*}^p[F].$$

Using (1), $H_{w^*}^p[F] = \overline{I_{w^*}[F]}$, so

$$C_\Phi g \in \overline{I_{w^*}[F]}.$$

(2) $\Rightarrow$ (3). This is immediate, since $\delta_{x_0} \in \text{FBL}^p[E] \subseteq I_{w^*}[E]$ and the rank-one maps Φ_f are admissible. For completeness, let us check admissibility. If $(y_j^*)_{j=1}^m \subset F^*$ has weak p -summing norm at most 1, then

$$\|(\Phi_f(y_j^*))_{j=1}^m\|_{p, \text{weak}} \leq \|x_0^*\| \left(\sum_{j=1}^m |f(y_j^*)|^p \right)^{1/p} \leq \|x_0^*\| \|f\|_{\text{FBL}^p[F]}.$$

Thus $\|\Phi_f\|_p < \infty$. Positive homogeneity and weak-star continuity on B_{F^*} follow from the same properties of f .

(3) $\Rightarrow$ (1). Take $f \in H_{w^*}^p[F]$. With $\Phi_f(y^*) = f(y^*)x_0^*$, we have

$$C_{\Phi_f}(\delta_{x_0})(y^*) = \delta_{x_0}(\Phi_f(y^*)) = f(y^*)x_0^*(x_0) = f(y^*)$$

for every $y^* \in F^*$. By (3), $f \in \overline{I_{w^*}[F]}$. Since f was arbitrary and the reverse inclusion $\overline{I_{w^*}[F]} \subseteq H_{w^*}^p[F]$ is Lemma 2.1(viii) of [1], we obtain $\overline{I_{w^*}[F]} = H_{w^*}^p[F]$. $\square$

A Positive Reflexive-BAP Case of Question 4.2

Theorem 2 (Reflexive BAP targets). *Let E be a reflexive real Banach space with the bounded approximation property. Then, for every $1 \leq p < \infty$,*

$$\overline{I_{w^*}[E]} = H_{w^*}^p[E],$$

where the closure is taken in the $\text{FBL}^p[E]$ -norm.

Proof. Fix $1 \leq p < \infty$. By Lemma 2.1(viii) of [1], it is enough to prove $H_{w^*}^p[E] \subseteq \overline{I_{w^*}[E]}$.

Let $f \in H_{w^*}^p[E]$. Since E is reflexive, Theorem 5.5 of [1] gives $H_{w^*}^p[E] = H^p[E]$. Proposition 7.7 of the source paper cited in [1] then gives a positive Radon probability measure μ on the weakly compact unit ball B_E such that

$$|f(x^*)| \leq \|f\|_{\text{FBL}^p[E]} f_\mu^p(x^*) \quad (x^* \in E^*),$$

where

$$f_\mu^p(x^*) = \left(\int_{B_E} |x^*(x)|^p d\mu(x) \right)^{1/p}.$$

Thus it suffices to prove $f_\mu^p \in \overline{I_{w^*}[E]}$, because $\overline{I_{w^*}[E]}$ is an ideal in $H_{w^*}^p[E]$.

By the Amir–Lindenstrauss theorem that Eberlein compacta have separable measure supports [2], the Radon measure μ is carried by a weakly separable compact subset of B_E . Mazur's theorem then implies that the closed linear span of this support is a norm-separable subspace $S \subseteq E$. Since E has the bounded approximation property, we can choose a sequence (T_n) of finite-rank operators on E with $\sup_n \|T_n\| \leq \lambda < \infty$ and $T_n x \rightarrow x$ in norm for every $x \in S$: take a countable dense set in S and diagonalize the uniformly bounded BAP net on its finite initial segments. For each n , set $F_n = T_n(E)$, and define $g_n : F_n^* \rightarrow \mathbb{R}$ by

$$g_n(u^*) = \left(\int_{B_E} |u^*(T_n x)|^p d\mu(x) \right)^{1/p}.$$

Since F_n is finite-dimensional, the finite-dimensional equality from Theorem 3.5 of [1] gives $g_n \in H_{w^*}^p[F_n] = \text{FBL}^p[F_n]$. If $J_n : F_n \hookrightarrow E$ denotes inclusion, then

$$h_n := C_{J_n^*} g_n \in \text{FBL}^p[E] \subseteq \overline{I_{w^*}[E]}.$$

Explicitly,

$$h_n(x^*) = \left(\int_{B_E} |x^*(T_n x)|^p d\mu(x) \right)^{1/p}.$$

We now show that $h_n \rightarrow f_\mu^p$ in the $\text{FBL}^p[E]$ -norm. Let $x_1^*, \dots, x_m^* \in E^*$ have weak p -summing norm at most 1. The reverse triangle inequality in $L_p(\mu)$ gives

$$|f_\mu^p(x_j^*) - h_n(x_j^*)| \leq \left(\int_{B_E} |x_j^*((I - T_n)x)|^p d\mu(x) \right)^{1/p}.$$

Therefore

$$\begin{aligned} \sum_{j=1}^m |f_\mu^p(x_j^*) - h_n(x_j^*)|^p &\leq \int_{B_E} \sum_{j=1}^m |x_j^*((I - T_n)x)|^p d\mu(x) \\ &\leq \int_{B_E} \|(I - T_n)x\|^p d\mu(x). \end{aligned}$$

For μ -almost every x , we have $x \in S$, so the integrand tends pointwise to zero and is bounded by $(1 + \lambda)^p$. Dominated convergence yields

$$\|f_\mu^p - h_n\|_{\text{FBL}^p[E]} \leq \left(\int_{B_E} \|(I - T_n)x\|^p d\mu(x) \right)^{1/p} \longrightarrow 0.$$

Thus $f_\mu^p \in \overline{I_{w^*}[E]}$, and the ideal domination above gives $f \in \overline{I_{w^*}[E]}$. $\square$

Corollary 1 (Universal equivalence). *For a fixed $1 \leq p < \infty$, Question 6.3 of [1] has an affirmative answer for all infinite-dimensional Banach spaces E, F if and only if Question 4.2 has an affirmative answer for every infinite-dimensional Banach space F .*

Proof. Apply Theorem 1 for each target space F . $\square$

Corollary 2 (Known positive target classes). *If F is an infinite-dimensional Banach space for which $\overline{I_{w^*}[F]} = H_{w^*}^p[F]$, then Question 6.3 is affirmative for this target F and arbitrary infinite-dimensional domain E . In particular this holds for $F = \ell_1(A)$, for any infinite set A , by the positive result for $\ell_1(A)$ proved in Section 4 of [1]. It also holds for every infinite-dimensional reflexive real Banach space with the bounded approximation property, for every $1 \leq p < \infty$, by Theorem 2. In particular it holds for Hilbert targets for every $1 \leq p < \infty$.*

Limitations

This packet does not solve Question 4.2 for all Banach spaces. Its first contribution is structural: Question 6.3 is equivalent to Question 4.2 target-by-target, and any future counterexample to Question 4.2 immediately yields a rank-one counterexample to Question 6.3. Its second contribution is the positive reflexive-BAP theorem. The remaining frontier includes reflexive spaces without the bounded approximation property and non-reflexive spaces beyond the $\ell_1(A)$ class covered by the source paper. Question 4.1 and the separable-quotient obstruction around Theorem 4.3 of the source paper are untouched.

Novelty and Literature Check

The source paper already observes that rank-one maps built from $f \in H_{w^*}^p[F] \setminus \text{FBL}^p[F]$ need not preserve FBL^p . The present argument uses the same rank-one mechanism at the level of the closed ideal $\overline{I_{w^*}}$, giving an exact equivalence between the two open questions. A local index search for arXiv:2404.06497 and the phrases I_{w^*} , $H_{w^*}^p$, C_Φ , and “positively homogeneous functions” found no earlier exact run packet for this paper before this packet was created. Web searches on 2026-06-23 and again on 2026-06-26 for the title, arXiv id, I_{w^*} , $H_{w^*}^p$, C_Φ , “positively homogeneous functions”, and related Hilbert/BAP phrases found only the source arXiv paper and no later resolution or reflexive-BAP treatment.

Human Review Recommendation

Review as a likely valid strengthened partial result. For the rank-one equivalence, check the use of Lemma 6.1(ii) of [1] in $(1) \Rightarrow (2)$ and the calculation of the p -norm of the rank-one map Φ_f . For the reflexive-BAP theorem, check the use of the Amir–Lindenstrauss separable-support theorem for Eberlein compacta, the compressed finite-dimensional functions g_n , the use of the finite-dimensional equality in Theorem 3.5 of [1], and the FBL^p -norm estimate controlled by $\int \|(I - T_n)x\|^p d\mu(x)$.

References

- [1] N. J. Laustsen and P. Tradacete, *Banach lattices of positively homogeneous functions induced by a Banach space*, arXiv:2404.06497, 2024.
- [2] D. Amir and J. Lindenstrauss, *The structure of weakly compact sets in Banach spaces*, Annals of Mathematics (2) 88 (1968), 35–46.